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## EMCH 574 – Vibration and Waves

### Homework 01

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## Assignment

- G signifies students that take this class for Graduate Credit; they must do all the items.
- UG students are not required to do the G work, but they may do it for bonus points.
- Problems solved beyond the minimum requirements will be given bonus points.
- Challenge problems are optional; they will earn bonus points.

## Notes

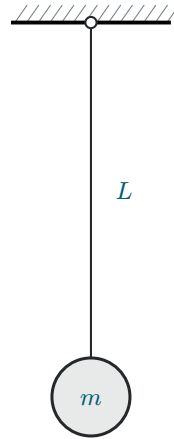
- Show your work to obtain partial credit if appropriate.
- Always display input data!
- Partial credit might be given based on work done and MATLAB codes if they are attached; without them, no partial credit can be given.
- Display numerical results to three significant digits (four if the first digit is 1).
- Use SI system of measurement [reference](#).
- Scale units using appropriate prefixes (e.g., nano, micro, milli, kilo, mega, giga, and so on) [reference](#).
- Problems solved beyond the minimum requirements will be given bonus points.
- Challenge problems are optional. They may be solved for extra credit.
- Normalized plots are plots with the *y*-axis scaled between  $-1$  and  $+1$ . A plotted variable is normalized by dividing it by its maximum absolute value.

## Notations and Abbreviations

- BVP = boundary value problem.
- EOM = equation of motion.
- FBD = free body diagram.
- FRF = frequency response function.
- IVP = initial value problem.
- LHS = left hand side.
- NLM = Newton's law of motion.
- ODE = ordinary differential equation.
- PDE = partial differential equation.
- RHS = right hand side.

# 1 Pendulum

## A.1 Analyze Pendulum



**Figure 1:** Pendulum of length  $L$  and mass  $m$ .

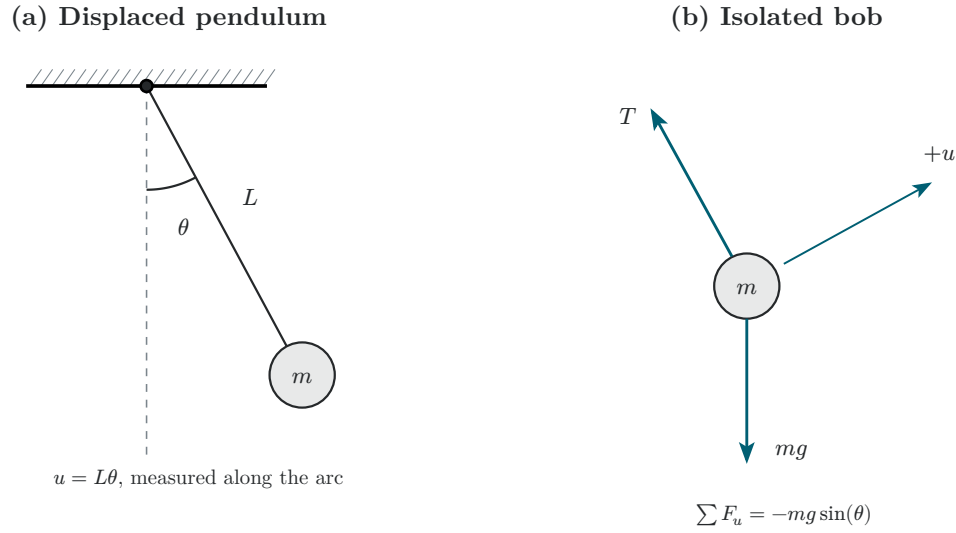
Consider a pendulum of length  $L$  and mass  $m$  as shown in Figure 1.

**Do the following.**

- (1) Draw the pendulum displaced by an angle  $\theta$  from the vertical.
- (2) Draw free-body diagram (FBD).
- (3) Write Newton's law of motion (NLM) equation.
- (4) Apply small-angle approximation and deduce the equation of motion (EOM). Clearly state your assumptions.
- (5) Deduce the governing equation.
- (6) Assume  $\theta(t) = e^{st}$  and substitute into the governing equation.
- (7) Deduce and solve the characteristic equation.
- (8) What type are the roots  $s_{1,2}$ ? Choose one of the following options:
  - (a) Real.
  - (b) Imaginary.
  - (c) Complex.
- (9) Define the natural frequency  $\omega_n$ .
- (10) Express the solutions  $s_1$  and  $s_2$  of the characteristic equation in terms of  $\omega_n$ .
- (11) Write the solution in terms of exponential functions of  $\omega_n t$ .
- (12) Write the solution in terms of trigonometric functions of  $\omega_n t$ .
- (13) Write the Hz frequency  $f_n$ .

- (14) Write the period of oscillation  $\tau$ .
- (15) Write the particle motion  $u(t)$  in terms of:
- (a) Exponential functions.
  - (b) Trigonometric functions.
- (16) Solve the initial value problem (IVP) and find the response  $u(t)$  for:
- (i) Initial displacement  $u_0$ .
  - (ii) Initial velocity  $\dot{u}_0$ .
  - (iii) Simultaneous application of both initial displacement  $u_0$  and initial velocity  $\dot{u}_0$ .

## Solution to A.1

**Figure 2:** Free-body diagram for Problem A.1.

Take positive  $\theta$  in the direction shown and let  $u = L\theta$  be the tangential arc displacement. Then  $\ddot{u} = L\ddot{\theta}$ . Assume a point mass, a massless inextensible rod or taut string of length  $L$ , a fixed frictionless pivot, uniform gravity, planar motion, and negligible air resistance.

The weight is  $mg$ , and the tension  $T$  has no tangential component. Therefore, the tangential force is the restoring component of gravity,

$$\sum F_u = -mg \sin \theta = m\ddot{u} = mL\ddot{\theta} \quad (\text{A.1.3})$$

For small angular displacement in radians,  $\sin \theta \approx \theta$ . Substituting this approximation gives the equation of motion

$$mL\ddot{\theta} + mg\theta = 0 \quad (\text{A.1.4})$$

Dividing by  $mL$  gives the governing equation

$$\ddot{\theta} + \frac{g}{L}\theta = 0 \quad (\text{A.1.5})$$

To solve the governing equation, assume  $\theta = e^{st}$ . Then  $\ddot{\theta} = s^2 e^{st}$ , so substitution gives

$$\left(s^2 + \frac{g}{L}\right) e^{st} = 0.$$

Because  $e^{st} \neq 0$ , the characteristic equation is

$$\boxed{s^2 + \frac{g}{L} = 0} \quad (\text{A.1.7})$$

Since  $g/L > 0$ , solving this equation gives purely imaginary roots. Therefore, the answer to item 8 is

$$\boxed{\text{Imaginary}} \quad (\text{A.1.8.b})$$

Define the natural angular frequency by  $\omega_n^2 = g/L$ . The natural frequency is therefore

$$\boxed{\omega_n = \sqrt{\frac{g}{L}}} \quad (\text{A.1.9})$$

Using this definition, the roots become

$$\boxed{s_1 = i\omega_n, \quad s_2 = -i\omega_n} \quad (\text{A.1.10})$$

The two roots produce the exponential angular response

$$\boxed{\theta(t) = D_1 e^{i\omega_n t} + D_2 e^{-i\omega_n t}} \quad (\text{A.1.11})$$

Using Euler's identity, the same response can be written as

$$\boxed{\theta(t) = A_\theta \cos(\omega_n t) + B_\theta \sin(\omega_n t)} \quad (\text{A.1.12})$$

Because  $\omega_n = 2\pi f_n$  and  $f_n = 1/\tau$ , the frequency and period are

$$\boxed{f_n = \frac{\omega_n}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{g}{L}}} \quad (\text{A.1.13})$$

$$\boxed{\tau = \frac{1}{f_n} = \frac{2\pi}{\omega_n} = 2\pi \sqrt{\frac{L}{g}}} \quad (\text{A.1.14})$$

Multiplying the angular response by  $L$  gives the particle-motion response. In exponential form, this is

$$\boxed{u(t) = C_1 e^{i\omega_n t} + C_2 e^{-i\omega_n t}} \quad (\text{A.1.15.a})$$

Euler's identity also gives the trigonometric forms

$$\boxed{u(t) = A \cos(\omega_n t) + B \sin(\omega_n t) = C \sin(\omega_n t + \varphi)} \quad (\text{A.1.15.b})$$

For real  $A$  and  $B$ , the amplitude and phase are  $C = \sqrt{A^2 + B^2}$  and  $\varphi = \text{atan2}(A, B)$ .

Differentiate the trigonometric response and evaluate it at  $t = 0$ . The initial conditions give

$$u(0) = A = u_0, \quad \dot{u}(0) = \omega_n B = v_0, \quad v_0 = \dot{u}_0.$$

With initial displacement only,  $v_0 = 0$ , so

$$\boxed{u(t) = u_0 \cos(\omega_n t)} \quad (\text{A.1.16.i})$$

With initial velocity only,  $u_0 = 0$ , so

$$\boxed{u(t) = \frac{v_0}{\omega_n} \sin(\omega_n t)} \quad (\text{A.1.16.ii})$$

With both initial conditions applied, the response is

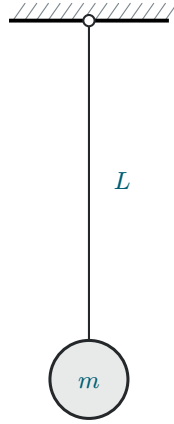
$$\boxed{u(t) = u_0 \cos(\omega_n t) + \frac{v_0}{\omega_n} \sin(\omega_n t)} \quad (\text{A.1.16.iii})$$

The corresponding exponential constants are

$$C_1 = \frac{1}{2} \left( u_0 - \frac{iv_0}{\omega_n} \right), \quad C_2 = \frac{1}{2} \left( u_0 + \frac{iv_0}{\omega_n} \right).$$

The mass cancels when the governing equation is divided by  $m$ , so the ideal small-angle pendulum frequency depends on  $g$  and  $L$ , but not on  $m$ .

## A.2 Calculate Pendulum Response to Initial Displacement



**Figure 3:** Pendulum of length  $L$  and mass  $m$ .

Take  $g = 9.81 \text{ m/s}^2$  as the generally accepted value of gravitational acceleration. Consider a pendulum of nominal length  $L_{\text{nominal}} = 565 \text{ mm}$  and mass  $m = 52 \text{ kg}$  as shown in Figure 3. When manufactured in series production, the pendulum lengths are not exact because of manufacturing tolerances. The lengths of the ten pendulum sets are

$$L = [561.535, 562.691, 563.845, 563.845, 565.000, \\ 565.000, 566.155, 566.155, 567.309, 568.464] \text{ mm}.$$

**Do the following.**

- (1) For the ten pendulum sets, calculate.
  - (a) Natural frequency  $\omega_n$  in rad/s.
  - (b) Hz natural frequency  $f_n$ .
  - (c) Period of oscillation  $\tau$ .
  - (d) Display in table format  $L, \omega_n, f_n, \tau$ .
- (2) Mean values: calculate and display the mean values of  $\omega_n, f_n, \tau$ .
- (3) Initial value problem (IVP). Assume initial displacement  $u_0 = 2 \text{ mm}$  and zero initial velocity. Use the mean values of  $\omega_n, f_n$ , and  $\tau$  to complete items (a)–(c).
  - (a) Calculate and display the position of the mass  $u(t_1)$  after  $t_1 = 10 \text{ s}$ .
  - (b) Calculate and display the duration  $t_{N_{\text{osc}}}$  of  $N_{\text{osc}}$  oscillations and the corresponding position of the mass  $u_{N_{\text{osc}}}$  for  $N_{\text{osc}} = 10$ .
  - (c) Plot the response of the system for  $N_{\text{osc}}$  oscillations and use a data tip to find  $u(t_1)$  at  $t_1$ .

**Solution to A.2**

Use SI units by converting each length from millimetres to metres. For set  $j$ , the pendulum relation from A.1 gives

$$\omega_{n,j} = \sqrt{\frac{g}{L_j}} \quad (\text{A.2.1.a})$$

The frequency and period follow from  $\omega_{n,j} = 2\pi f_{n,j}$  and  $f_{n,j} = 1/\tau_j$ , so

$$f_{n,j} = \frac{\omega_{n,j}}{2\pi} \quad (\text{A.2.1.b})$$

$$\tau_j = \frac{2\pi}{\omega_{n,j}} \quad (\text{A.2.1.c})$$

Using  $g = 9.81 \text{ m/s}^2$  and the ten lengths gives the following values for A.2.1.d.

| Set | $L$ (mm) | $\omega_n$ (rad/s) | $f_n$ (Hz) | $\tau$ (s) |
|-----|----------|--------------------|------------|------------|
| 1   | 561.535  | 4.18               | 0.665      | 1.503      |
| 2   | 562.691  | 4.18               | 0.665      | 1.505      |
| 3   | 563.845  | 4.17               | 0.664      | 1.506      |
| 4   | 563.845  | 4.17               | 0.664      | 1.506      |
| 5   | 565.000  | 4.17               | 0.663      | 1.508      |
| 6   | 565.000  | 4.17               | 0.663      | 1.508      |
| 7   | 566.155  | 4.16               | 0.663      | 1.509      |
| 8   | 566.155  | 4.16               | 0.663      | 1.509      |
| 9   | 567.309  | 4.16               | 0.662      | 1.511      |
| 10  | 568.464  | 4.15               | 0.661      | 1.513      |

Compute the mean of each calculated column using  $\bar{q} = \frac{1}{10} \sum_{j=1}^{10} q_j$ . The mean values required in A.2.2 are

$$\overline{\omega_n} = 4.17 \text{ rad/s} \quad (\text{A.2.2})$$

$$\overline{f_n} = 0.663 \text{ Hz} \quad (\text{A.2.2})$$

$$\overline{\tau} = 1.508 \text{ s} \quad (\text{A.2.2})$$

For A.2.3, the initial velocity is zero, so the A.1 response reduces to

$$u(t) = u_0 \cos(\overline{\omega_n} t) = 2 \cos(4.166893t) \text{ mm} \quad (\text{A.2.3})$$

At  $t_1 = 10$  s, substitution into the response gives

$$u(t_1) = 2 \cos(4.166893 \times 10) \text{ mm} = -1.352 \text{ mm} \quad (\text{A.2.3.a})$$

For ten oscillations, multiply the mean period by  $N_{\text{osc}} = 10$ . The duration and corresponding position are

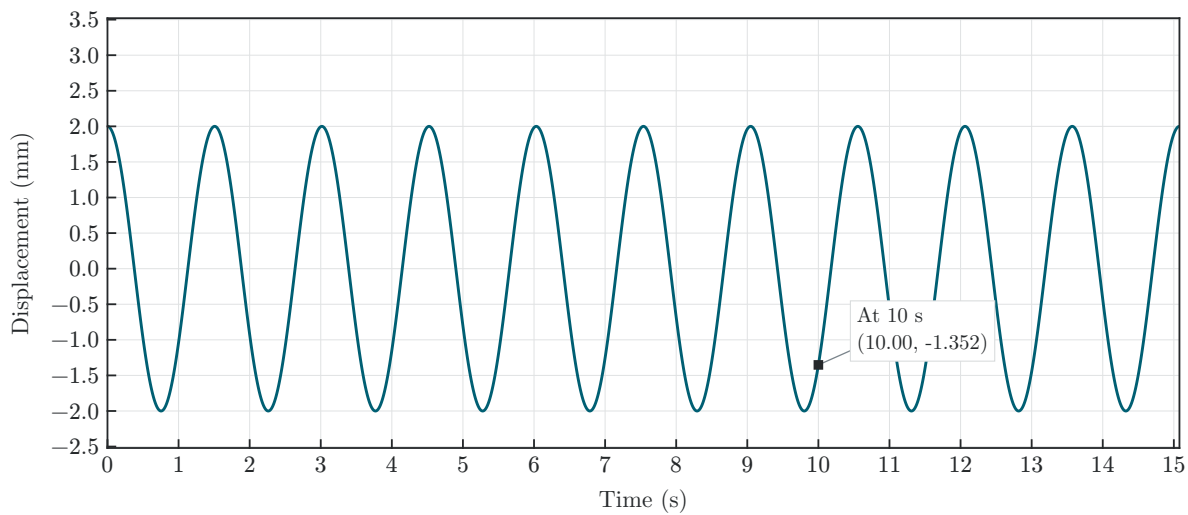
$$t_{N_{\text{osc}}} = 10\bar{\tau} = 15.08 \text{ s} \quad (\text{A.2.3.b})$$

$$u_{N_{\text{osc}}} = u(10\bar{\tau}) = 2.000 \text{ mm} \quad (\text{A.2.3.b})$$

Plot the response over  $0 \leq t \leq 10\bar{\tau}$  using the boxed A.2.3 expression. The figure below shows the ten-oscillation response and marks the value at  $t_1 = 10$  s. The plotted value is

$$u(10) = -1.352 \text{ mm} \quad (\text{A.2.3.c})$$

## A2. Pendulum released from 2 mm

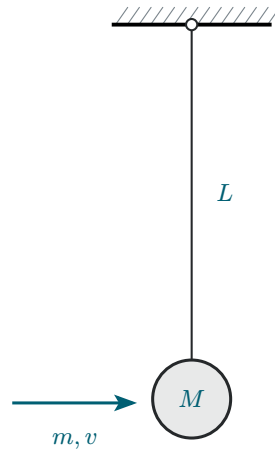


**Figure 4:** A.2.3.c. Pendulum response over ten oscillations with a data tip at  $t = 10$  s.

A MATLAB script that reproduces the plot is

```
01  u0 = 2;  
02  wn = 4.166893;  
03  t1 = 10;  
04  Nosc = 10;  
05  
06  tau = 2*pi/wn;  
07  t = linspace(0,Nosc*tau,1000);  
08  u = u0*cos(wn*t);  
09  
10  plot(t,u,t1,u0*cos(wn*t1),'ks');  
11  xlabel('Time (s)');  
12  ylabel('u (mm)');
```

### A.3 Calculate Pendulum Response to Impact



**Figure 5:** *Impact excitation of a pendulum system.*

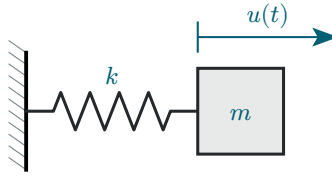
Take  $g = 9.81 \text{ m/s}^2$  as the generally accepted value of gravitational acceleration. Consider a pendulum of nominal length  $L = 565 \text{ mm}$  and mass  $M = 52 \text{ kg}$  as shown in Figure 5. The pendulum is initially at rest hanging vertically. A small projectile of mass  $m = 22 \text{ g}$  and velocity  $v = 945 \text{ m/s}$  hits the mass  $M$  and remains embedded into it. This impact imparts an initial velocity to the system that starts to oscillate.

**Do the following.**

- (1) Display input data.
- (2) Calculate the rad/sec natural frequency  $\omega_n$  and period of oscillation  $\tau$  of the pendulum with projectile embedded into it.
- (3) Calculate the initial velocity of the system  $\dot{u}_0$  as a result of the interaction between the projectile and the mass.
- (4) Calculate the response  $u(t_1)$  after  $t_1 = 10 \text{ s}$ .
- (5) Plot the response of the system for  $N_{\text{osc}} = 10$  oscillations and find on the plot:
  - (a) The largest response.
  - (b) The response value calculated at item (4).
- (6) Discuss your results.

## 2 Spring-Mass-Damper Vibrating Systems

### B.1 Analyze Spring-Mass Vibrating System



**Figure 6:** Spring-mass vibrating system.

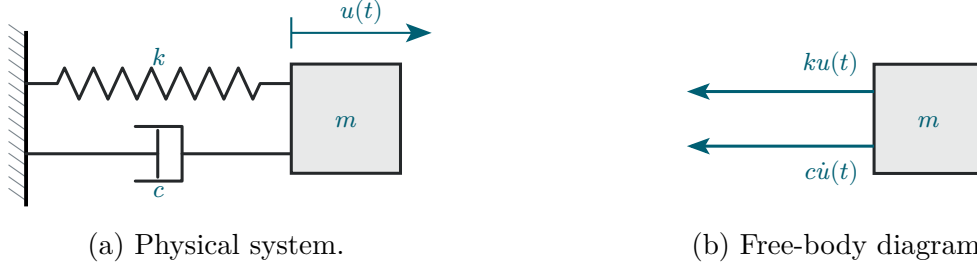
Consider a spring-mass vibrating system as shown in Figure 6.

**Do the following.**

- (1) Draw free-body diagram (FBD).
- (2) Write NLM equation.
- (3) Deduce the equation of motion (EOM).
- (4) Deduce the governing equation. Clearly state your assumptions.
- (5) Assume  $u(t) = e^{st}$  and substitute into the governing equation.
- (6) Deduce the characteristic equation.
- (7) Define the natural frequency  $\omega_n$  in terms of  $m, k$ .
- (8) Solve the characteristic equation and find  $s_{1,2}$  in terms of  $\omega_n$ .
- (9) What type are the roots  $s_{1,2}$ ? Choose one of the following options:
  - (a) Real.
  - (b) Imaginary.
  - (c) Complex.
- (10) Write the solution in terms of exponential functions.
- (11) Write the solution in terms of trigonometric functions.
- (12) Solve the initial value problem (IVP) and find the response  $u(t)$  for:
  - (i) Initial displacement  $u_0$ .
  - (ii) Initial velocity  $\dot{u}_0$ .
  - (iii) Simultaneous application of both initial displacement  $u_0$  and initial velocity  $\dot{u}_0$ .

## B.2 Analyze Spring-Mass-Damper Vibrating System

Consider the 1-dof damped system consisting of a spring,  $k$ , mass,  $m$ , and viscous damper,  $c$ , as presented in Figure 7. Assume the mass  $m$  is displaced from the equilibrium position by a time-dependent displacement  $u(t)$ .



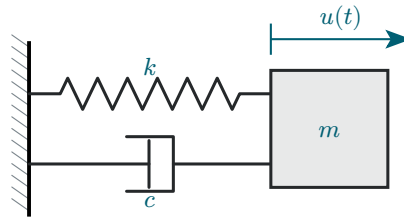
**Figure 7:** Mass-spring-damper vibrating system.

**Do the following.**

- (1) State your assumptions in performing this analysis.
- (2) Draw free-body diagram (FBD) and deduce the sum of forces in the  $x$  direction.
- (3) Write NLM equation.
- (4) Deduce the equation of motion (EOM) as an ODE in terms of  $m$ ,  $c$ ,  $k$ .
- (5) Assume  $u(t) = e^{st}$  and substitute into EOM.
- (6) Deduce the characteristic equation.
- (7) Find the roots  $s_{1,2}$  of the characteristic equation.
- (8) Define critical damping  $c_{cr}$  in terms of  $m$  and  $k$ .
- (9) Define damping ratio  $\zeta$ .
- (10) Discuss the values of  $\zeta$  for the three damping cases:
  - (a) Underdamped.
  - (b) Critically damped.
  - (c) Overdamped.
- (11) Express the physical damping  $c$  in terms of damping ratio  $\zeta$ , mass  $m$ , and stiffness  $k$ .
- (12) Define the natural frequency  $\omega_n$  in terms of  $m$  and  $k$ .
- (13) Express in terms of  $\omega_n$ ,  $\zeta$  the  $s_{1,2}$  roots found at (7).
- (14) Assume underdamped system and express the  $s_{1,2}$  roots of (13) as complex numbers in terms of  $\omega_n$  and  $\zeta$ .
- (15) Define the damped frequency  $\omega_d$ .
- (16) Substitute (15) into (14) and express the roots  $s_1$  and  $s_2$  in terms of  $\omega_n$ ,  $\omega_d$ ,  $\zeta$ .
- (17) Assume underdamped conditions and state the inequality that  $\zeta$  should satisfy.

- (18) Write the general solution of the ODE deduced at (4) in terms of constants  $C_1$  and  $C_2$  multiplying exponential functions of  $s_1$  and  $s_2$ .
- (19) Use (16) to write the solution (18) in terms of exponential functions of  $\omega_n$ ,  $\omega_d$ ,  $\zeta$ .
- (20) Rewrite the solution (19) in terms of an exponential decay function multiplying the sum of two complex exponentials.
- (21) Rewrite the solution (20) in terms of an exponential decay function multiplying a trigonometric function and some constants  $C$  and  $\phi$  to be determined from the initial conditions.
- (22) Sketch the functions discussed at (21).
- (23) Solve the initial value problem (IVP) for  $\zeta < 1$  and find the response  $u(t)$  for initial displacement  $u_0$  with zero initial velocity.
- (24) Solve the initial value problem (IVP) and find the response  $u(t)$  for simultaneous application of initial displacement  $u_0$  and initial velocity  $\dot{u}_0$ .

### B.3 Calculate Spring-Mass-Damper Response to Initial Displacement



**Figure 8:** *Spring-mass-damper vibrating system.*

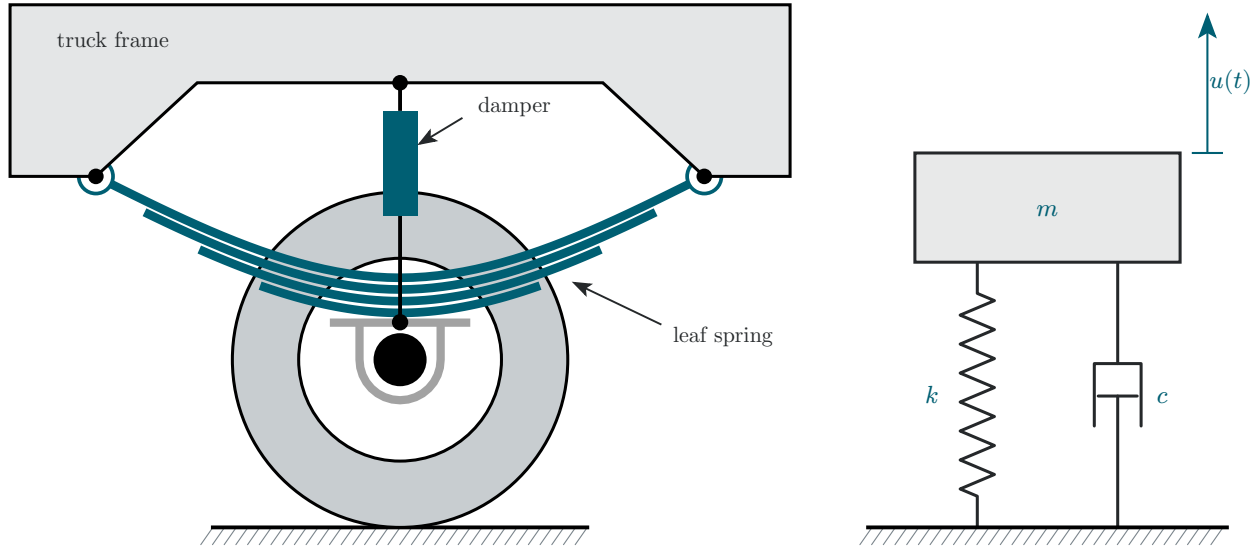
Consider a spring-mass vibrating system with  $m = 12 \text{ kg}$ ,  $k = 655 \text{ N/m}$ ,  $c = 19 \text{ N/(m/s)}$  as shown in Figure 8. The vibrating system is given an initial displacement  $u_0 = 3.5 \text{ mm}$  with zero initial velocity.

**Do the following.**

- (1) Display input data.
- (2) Calculate the natural frequency in rad/sec  $\omega_n$  and in Hz  $f_n$ .
- (3) Calculate the critical damping  $c_{cr}$ .
- (4) Calculate the damping ratio  $\zeta$ .
- (5) Calculate the damped frequency in rad/sec  $\omega_d$  and in Hz  $f_d$ .
- (6) Calculate the damped period of oscillation  $\tau_d$ .
- (7) Calculate the characteristic-equation roots  $s_1$ ,  $s_2$  and the constants  $C_1$ ,  $C_2$ .
- (8) Calculate the constants  $A$ ,  $B$  and then the constants  $C$ ,  $\phi$ .
- (9) Calculate the duration  $t_1$  of one oscillation and the corresponding position  $u_1 = u(t_1)$ .
- (10) Calculate the value  $u_{0.02}$  representing 2% of the initial displacement.
- (11) Plot the response of the system for  $N_{osc} = 10$  oscillations and use datatips on the plot to find the following:
  - (a) The largest response.
  - (b) The response value calculated at item (9).
  - (c) The time when the response magnitude has decreased permanently below  $u_{0.02}$  calculated at item (10).
- (12) Discuss your results.

### B.4 Calculate Truck Suspension Response

Consider a pickup truck wheel suspension consisting of a leaf spring and an oil damper. The truck and its suspension can be modeled, in first approximation, as a  $k$ - $m$ - $c$  spring-mass-damper system as shown in Figure 9.



**Figure 9:** Truck suspension consisting of a leaf spring and an oil damper. The wheel contact is treated as the road datum in the one-degree-of-freedom idealization.

The leaf spring can be approximated by a uniform beam of length  $L = 830 \text{ mm}$  and cross-section dimensions  $h = 3 \text{ mm}$  and  $b = 65 \text{ mm}$ , made of steel with  $E = 200 \text{ GPa}$ . The truck mass apportioned to the wheel can be estimated as  $m = 457 \text{ kg}$ . Assume initial displacement  $u_0 = 125 \text{ mm}$  and zero initial velocity  $\dot{u}_0 = 0$ .

**Do the following.**

- (1) Display input data.
- (2) Calculate the second moment of area  $I$  of the beam cross section.
- (3) Calculate the flexural stiffness  $EI$  of the beam.
- (4) Calculate the spring constant  $k$  of the system.
- (5) Calculate the natural frequency in rad/sec  $\omega_n$  and in Hz  $f_n$ .
- (6) Calculate the critical damping  $c_{cr}$ .
- (7) Assume a 25% damping ratio, i.e.,  $\zeta = 0.25$ , and calculate the physical damping  $c$ . Print  $\zeta$  and  $c$ .
- (8) Calculate the damped frequency in rad/sec  $\omega_d$  and in Hz  $f_d$ .
- (9) Calculate the damped period of oscillation  $\tau_d$ .
- (10) Calculate the characteristic-equation roots  $s_1, s_2$  and the constants  $C_1, C_2$ .
- (11) Calculate the constants  $A, B$  and then the constants  $C, \phi$ .

- 
- (12) Calculate the duration  $t_1$  of one oscillation and the corresponding position  $u_1 = u(t_1)$ .
- (13) Calculate the value  $u_{0.02}$  representing 2% of the initial displacement.
- (14) Plot the response of the system for  $N_{\text{osc}} = 10$  oscillations and use datatips on the plot to find the following:
- (a) The largest response.
  - (b) The response value calculated at item (12).
  - (c) The time when the response magnitude has decreased permanently below  $u_{0.02}$  calculated at item (13).
- (15) Discuss your results.
- (16) Graduate work. Assume the damper is worn out such that  $\zeta$  is reduced to  $\zeta = 10\%$ .
- (a) Repeat the calculation of items (6) through (12).
  - (b) Compare and discuss the two results.

## B.5 Overdamped Spring-Mass-Damper Systems

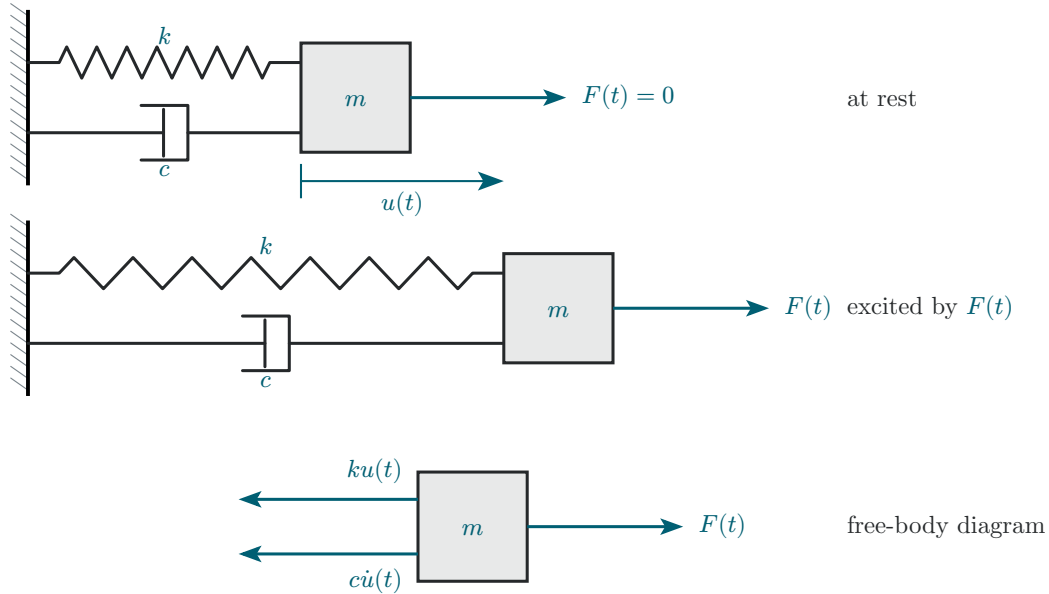
Consider a spring-mass-damper system with natural frequency  $f_n = 6.25 \text{ Hz}$  and various values of the damping ratio  $\zeta$ . The system is given an initial displacement  $u_0 = 13 \text{ mm}$  with zero initial velocity.

**Do the following.**

- (1) Display input data.
- (2) Calculate the rad/s frequency  $\omega_n$ .
- (3) Assume overdamped conditions  $\zeta > 1$  and develop the solution as follows:
  - (a) Write the expressions of the roots  $s_1, s_2$  in terms of  $\omega_n$  and  $\zeta$ .
  - (b) Write the solution  $u(t)$  in terms of  $s_1, s_2$  and constants  $C_1, C_2$ .
  - (c) Write the expression of the constants  $C_1, C_2$  in terms of initial displacement  $u_0$  and initial velocity  $\dot{u}_0$ .
  - (d) Take  $\zeta = 2.3$  and calculate the numerical values of  $s_1, s_2, C_1$ , and  $C_2$ .
  - (e) Plot the response  $u(t)$  for  $t = 0, \dots, 3 \text{ s}$ .
- (4) Assume critically damped conditions  $\zeta = 1$  and complete items (a)–(e).
  - (a) Write the expressions of the two partial solutions  $u_1(t), u_2(t)$  in terms of  $\omega_n$ .
  - (b) Write the expression of the complete solution in terms of  $\omega_n$  and  $C_1, C_2$ .
  - (c) Write the expressions of the constants  $C_1, C_2$  in terms of initial displacement  $u_0$  and initial velocity  $\dot{u}_0$ .
  - (d) Calculate the numerical values of  $C_1, C_2$ .
  - (e) Plot the response  $u(t)$  for  $t = 0, \dots, 3 \text{ s}$ .
- (5) Graduate work. Derive the ODE for  $\zeta = 1$  and verify that the two solutions  $u_1(t)$  and  $u_2(t)$  from (4)(a) satisfy the ODE.
- (6) Graduate work. Some race-car manufacturers design for critical damping, whereas others use an overdamped design. Discuss the advantages and disadvantages of the two options.

## B.6 Analyze Spring-Mass-Damper Forced Vibration

Consider a mass-spring-damper vibrating system excited by a time-dependent force  $F(t)$  as shown in Figure 10.



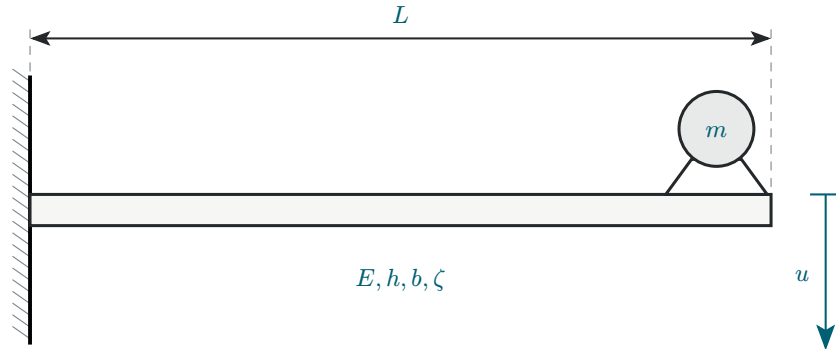
**Figure 10:** Spring-mass-damper system excited by force  $F(t)$ .

**Do the following.**

- (1) State your assumptions in performing this analysis.
- (2) Draw free-body diagram (FBD) and deduce the sum of forces in the  $x$  direction.
- (3) Write NLM equation.
- (4) Deduce the equation of motion (EOM) as an ODE in terms of  $m$ ,  $c$ ,  $k$ ,  $F(t)$ .
- (5) Assume harmonic excitation  $F(t) = \hat{F} e^{i\omega t}$ , substitute in (4), and divide by  $m$ .
- (6) Define the normalized excitation amplitude  $\hat{f} = \hat{F}/m$  and recall the definitions of  $\omega_n$  and  $\zeta$  to obtain a nonhomogeneous linear ODE in  $\omega_n$ ,  $\zeta$ ,  $\hat{f}$ .
- (7) Write the general solution  $u(t)$  as a sum of a complementary solution  $u_c(t)$  and a particular solution  $u_p(t)$ .
- (8) Write the general form of the complementary solution  $u_c(t)$ .
- (9) Find the particular solution  $u_p(t)$ .
- (10) Use (8), (9) to write the complete solution  $u(t)$ ; discuss the meaning of transient response and steady-state response.
- (11) Find an expression for the steady-state response amplitude  $\hat{u}_{ss}$  in terms of  $\omega_n$ ,  $\zeta$ ,  $\hat{f}$ , and  $\omega$ .
- (12) Develop the expression for the frequency response function  $\text{FRF}(\omega)$  in terms of  $\omega_n$ ,  $\zeta$ ,  $\omega$ .
- (13) Discuss the variation with frequency of FRF magnitude and phase.

- (14) Give the definition of the magnification factor  $M(\omega)$ .
- (15) Give the expression for the magnification factor  $M(\omega)$  in terms of  $\omega_n$  and  $\zeta$ .
- (16) Give the definition of the resonance frequency  $\omega_r$ .
- (17) Give the expression for the resonance frequency  $\omega_r$  in terms of  $\omega_n$  and  $\zeta$ .
- (18) Give the expression of the peak magnification  $M_r$ .
- (19) Give the expression for the FRF phase  $\phi(p)$ , where  $p = \omega/\omega_n$ .
- (20) Give the definition of phase resonance and deduce the FRF expression at phase resonance.
- (21) Calculate the magnification factor at phase resonance.
- (22) Calculate the magnitude of the physical response at phase resonance.
- (23) Discuss the difference between true resonance and phase resonance.
- (24) Write and compare the expressions for the resonance frequency  $\omega_r$  and damped frequency  $\omega_d$ .
- (25) Establish a formula to calculate the RPM corresponding to a rad/sec frequency  $\omega$ .

### B.7 Calculate Flex-Beam Response to Slightly Unbalanced Electric Motor Excitation



**Figure 11:** Damped cantilever beam excited by force  $F(t)$  from slightly unbalanced electric motor.

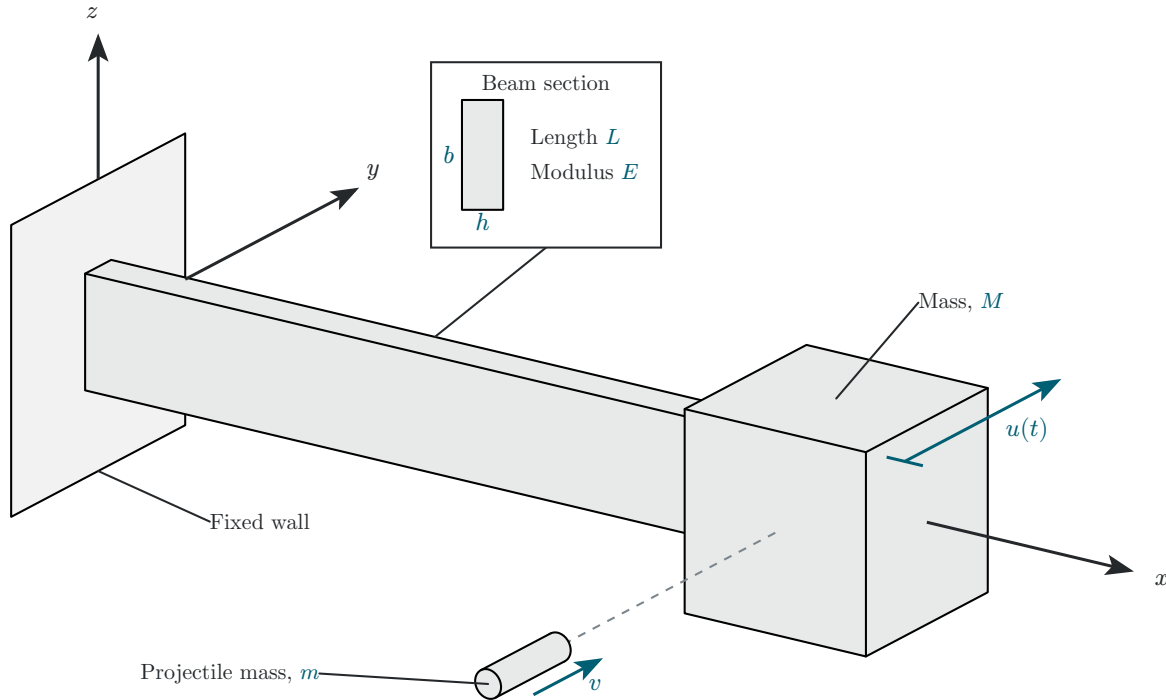
Consider a cantilever beam of length  $L = 568 \text{ mm}$ , elastic modulus  $E = 200 \text{ GPa}$ , cross-section dimensions  $h = 3 \text{ mm}$  and  $b = 37 \text{ mm}$ , and damping ratio  $\zeta = 14\%$ . (The damping is achieved through a proprietary damping treatment applied on the beam surface.) An electric motor of mass  $m = 0.52 \text{ kg}$  operating at  $1,500 \text{ RPM}$  is attached at the beam tip as shown in Figure 11. The electric motor is slightly unbalanced, producing a slight vibratory excitation  $F(t) = \hat{F}e^{i\omega t}$  with  $\hat{F} = 12 \text{ mN}$ . The excitation frequency  $\omega$  depends on the electric motor speed.

**Do the following.**

- (1) Display input data.
- (2) Calculate:
  - (a) Cross-section second moment of area  $I$ , flexural stiffness  $EI$ , and spring constant  $k$ .
  - (b) Natural frequency  $\omega_n$  rad/s, critical damping  $c_{cr}$ , physical damping  $c$ .
  - (c) Quasi-static deflection  $u_{qst}$  corresponding to  $\omega_n \rightarrow 0$ .
- (3) Calculate the values of:
  - (a) Resonant frequency  $\omega_r$  and damped frequency  $\omega_d$ .
  - (b) Peak magnification  $M_r$ .
  - (c) Magnification at phase resonance  $M_{90}$ .
  - (d) Physical response  $\hat{u}_{90}$  at phase resonance and its magnitude  $|\hat{u}_{90}|$ .
  - (e) Calculate the Hz frequencies  $f_r$ ,  $f_n$  and the corresponding RPM values.
- (4) Assume excitation at phase resonance:
  - (a) Calculate the excitation frequency  $\omega$  in rad/s and  $f$  in Hz.
  - (b) Calculate period of oscillation  $\tau$ .
  - (c) Plot the time response for  $N_{osc} = 10$  oscillations. Hint: plot the real part of the complex displacement  $u(t)$ .
- (5) Graduate work. Complete items (a)–(d).

- (a) Calculate and display the operational frequency  $f_0$  in Hz and the corresponding  $\omega_0$  in rad/sec. Plot the magnitude and phase of the frequency response function FRF vs. excitation  $f$  in Hz from quasi-static to  $f_0$  with  $N_f = 5000$  points.
- (b) Calculate and display the FRF magnitude at phase-resonance RPM, at double the phase resonance RPM, and at operational RPM. Display the corresponding Hz frequencies. Put the corresponding data tips on the magnitude plot.
- (c) Calculate the vibration magnitude  $|u_0|$  when the electric motor runs at its operational RPM.
- (d) Comment on what vibration would be observed as the motor is switched on and accelerates to reach its operational RPM.

### B.8 Graduate Work. Calculate Horizontal Flex-Beam Response to Impact



**Figure 12:** Impact excitation of a horizontal flex-beam vibration system.

Consider a cantilever beam of length  $L = 568 \text{ mm}$  with a mass  $M = 53 \text{ kg}$  attached at its tip as shown in Figure 12. The beam material is steel with  $E = 200 \text{ GPa}$ . The beam cross-section dimensions are  $h = 3 \text{ mm}$ ,  $b = 37 \text{ mm}$ . The beam is initially at rest. A small projectile of mass  $m = 18 \text{ g}$  and velocity  $v = 935 \text{ m/s}$  hits the mass and remains embedded in it. This impact imparts an initial velocity to the system that starts to oscillate with motion  $u(t)$  contained in the horizontal plane.

**Do the following.**

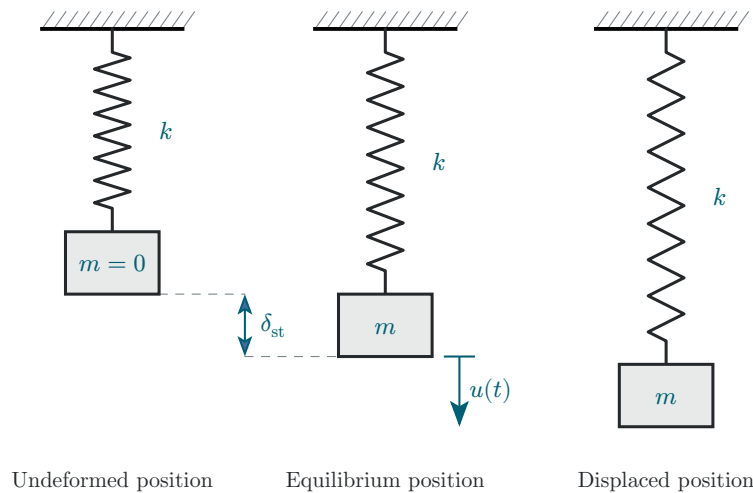
- (1) Display input data.
- (2) Calculate the second moment of area  $I$  for flexing consistent with the direction of motion  $u(t)$ .
- (3) Calculate the cross-sectional stiffness  $EI$  of the beam.
- (4) Calculate the flex-beam spring constant  $k$ .
- (5) Calculate the initial velocity of the system  $\dot{u}_0$  as a result of the interaction between the projectile and the mass.
- (6) Calculate the rad/sec natural frequency  $\omega_n$  and period of oscillation  $\tau$ .
- (7) Calculate the position of the mass  $u(t_1)$  after  $t_1 = 10 \text{ s}$ .
- (8) Plot the response of the system for  $N_{\text{osc}} = 10$  oscillations and find on the plot:
  - (a) The largest response.

- (b) The response calculated at item (7).
- (9) Discuss your results.

### 3 Extra Credit Challenge Problems

Challenge problems are optional.

#### C.1 Analyze Vertical Spring-Mass Vibration System



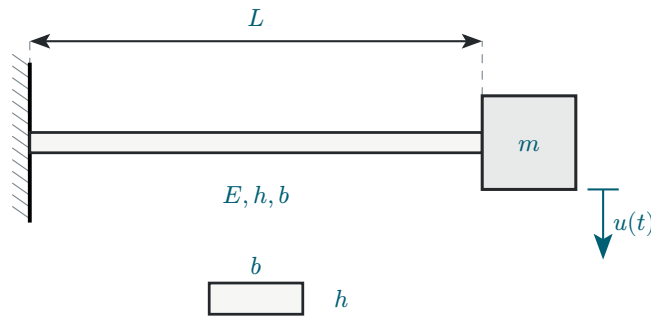
**Figure 13:** Vertical spring-mass vibration system.

Consider a vertical spring-mass vibration system consisting of a particle of mass  $m$  hanging on an elastic spring of stiffness  $k$  as shown in Figure 13.

**Do the following.**

- (1) Assume static equilibrium.
  - (a) Calculate the weight  $W$  acting on the mass  $m$ .
  - (b) Draw free-body diagram (FBD).
  - (c) Write equilibrium equation.
  - (d) Solve for the static displacement  $\delta_{st}$ .
- (2) Assume vibration with time-varying vertical displacement  $u(t)$ .
  - (a) Draw free-body diagram (FBD).
  - (b) Deduce equation of motion (EOM).
  - (c) Compare the EOM deduced here with the EOM derived for horizontal spring-mass vibration system of Problem B.1.
- (3) Graduate work. Comment on how the EOM would change in the presence of an arbitrary force field acting at an angle  $\theta$  with respect to the vertical axis.

## C.2 Analyze Vertical Flex-Beam Vibration



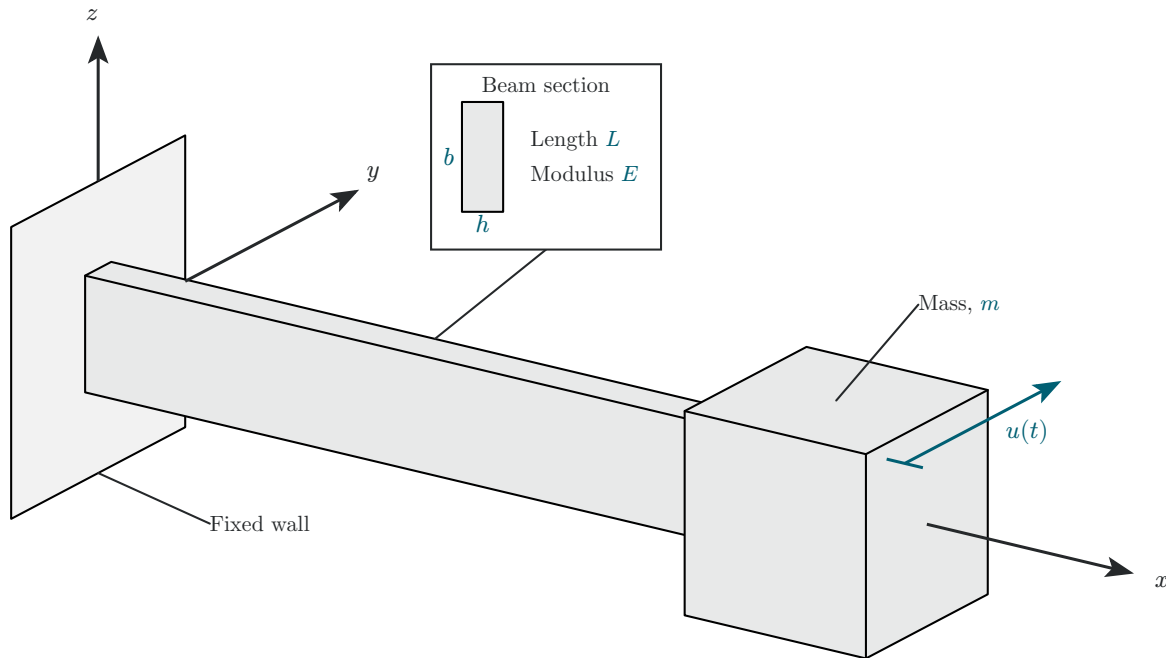
**Figure 14:** Vertical flex-beam vibrating system.

Consider a cantilever beam of length  $L$  with a mass  $m$  attached at its tip as shown in Figure 14. The beam cross-section properties are  $E$ ,  $h$ ,  $b$ .

**Do the following.**

- (1) Calculate the cross-sectional stiffness  $EI$  of the beam for vertical flexural deformation.
- (2) Calculate the spring constant  $k$  of the system.
- (3) Assume static equilibrium:
  - (a) Calculate the weight  $W$  acting on the mass  $m$ .
  - (b) Draw free-body diagram (FBD).
  - (c) Write equilibrium equation.
  - (d) Solve for the static displacement  $\delta_{st}$ .
- (4) Assume vibration with time-varying vertical displacement  $u(t)$  measured from the static equilibrium position:
  - (a) Draw free-body diagram (FBD).
  - (b) Deduce equation of motion (EOM).
  - (c) Compare the EOM deduced here with the EOM derived for horizontal flex-beam vibration system of Problem C.4.
- (5) Graduate work. Comment on how the EOM would change in the presence of an arbitrary force field acting at an angle  $\theta$  with respect to the vertical axis.

### C.3 Analyze Horizontal Flex-Beam Vibration



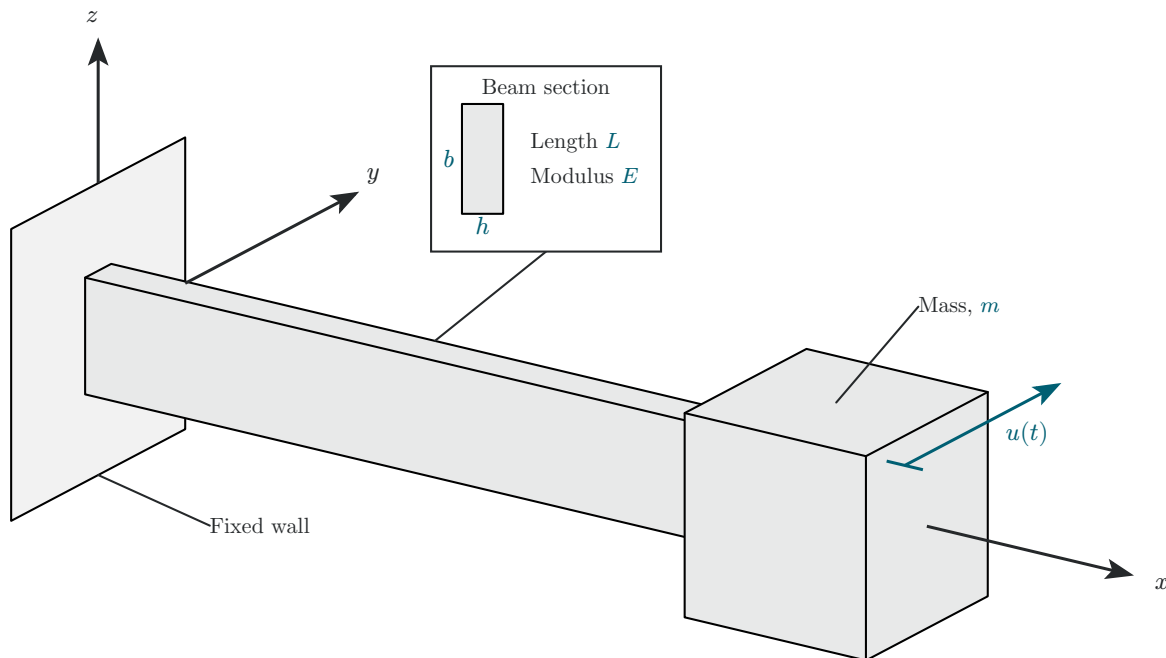
**Figure 15:** *Horizontal flex-beam vibrating system.*

Consider a cantilever beam of length  $L$  with a mass  $m$  attached at its tip as shown in Figure 15. The beam cross-section properties are  $E$ ,  $h$ ,  $b$ .

**Do the following.**

- (1) Assume vibration with horizontal time-varying displacement  $u(t)$  as shown in Figure 15.
- (2) Calculate the cross-sectional stiffness  $EI$  of the beam for flexural deformation consistent with  $u(t)$ .
- (3) Calculate the flex-beam spring constant  $k$ .
- (4) Draw free-body diagram (FBD) including all the forces acting on the mass  $m$ .
- (5) Deduce equation of motion (EOM).
- (6) Compare the EOM deduced here with the EOM derived in Problem B.1.
- (7) Solve the initial value problem (IVP) and find the response for simultaneous application of both initial displacement  $u_0$  and initial velocity  $\dot{u}_0$ .

### C.4 Calculate Horizontal Flex-Beam Response to Initial Displacement



**Figure 16:** *Flex-beam vibrating system.*

Consider a cantilever beam of length  $L = 568 \text{ mm}$  with a mass  $m = 5.35 \text{ kg}$  attached at its tip as shown in Figure 16. The beam material is steel with  $E = 200 \text{ GPa}$ . The beam cross-section dimensions are  $h = 3 \text{ mm}$ ,  $b = 37 \text{ mm}$ . The tip mass is given an initial displacement  $u_0 = 5.4 \text{ mm}$  with zero initial velocity.

**Do the following.**

- (1) Display input data.
- (2) Calculate the second moment of area  $I$  for flexing consistent with the direction of motion  $u(t)$ .
- (3) Calculate the cross-sectional stiffness  $EI$  of the beam.
- (4) Calculate the spring constant  $k$  of the system.
- (5) Calculate the natural frequency in rad/sec  $\omega_n$ .
- (6) Calculate the natural frequency in Hz  $f_n$ .
- (7) Calculate the period of oscillation in sec  $\tau$ .
- (8) Calculate the position of the mass  $u(t_1)$  after  $t_1 = 10 \text{ s}$ .
- (9) Plot the response of the system for  $N_{\text{osc}} = 25$  oscillations and use a data tip to find  $u(t_1)$  at  $t_1$ .
- (10) Discuss your results.

### C.5 Analyze Spring-Mass Forced Vibration

Consider a spring-mass vibrating system excited by a time-dependent force  $F(t)$  as shown in Figure 17.

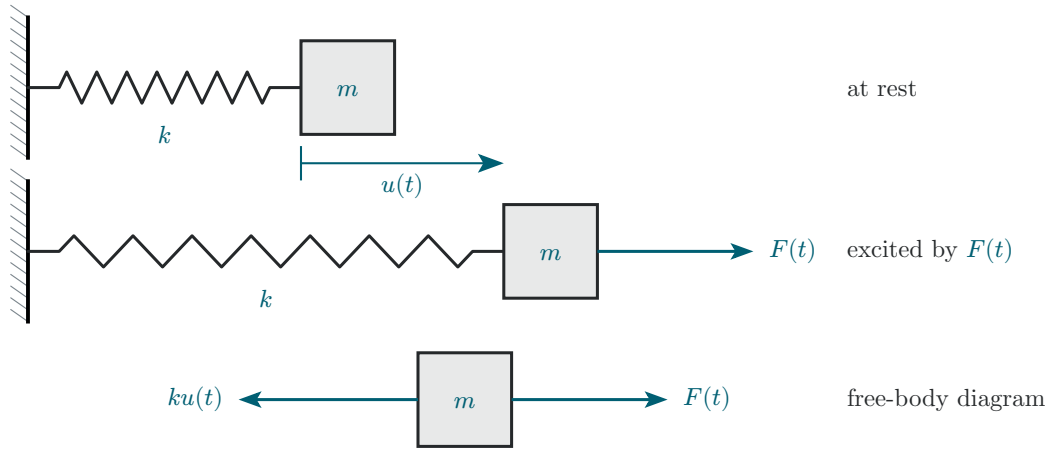


Figure 17: Spring-mass system excited by force  $F(t)$ .

### General Analysis of Spring-Mass Forced Vibration

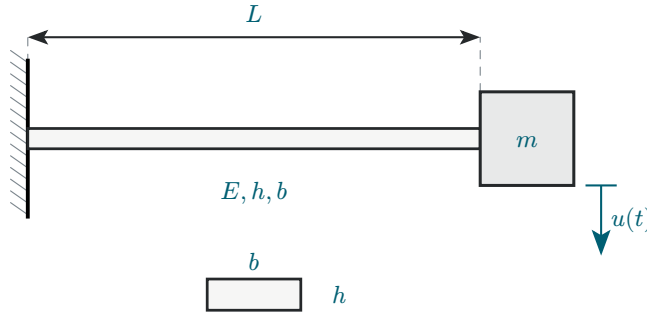
Do the following.

- (1) State your assumptions in performing this analysis.
- (2) Draw free-body diagram (FBD) and deduce the sum of forces in the  $x$  direction.
- (3) Write NLM equation.
- (4) Deduce the equation of motion (EOM) and express it as a nonhomogeneous ODE in terms of  $m$ ,  $k$ ,  $F(t)$ .
- (5) Write the ODE general solution as the superposition of a complementary solution  $u_c(t)$  and a particular solution  $u_p(t)$ .
- (6) Recall the complementary solution  $u_c(t)$  as the solution of the homogeneous equation derived in Problem B.1 and write it in terms of trigonometric functions multiplied by constants  $A$  and  $B$ .

### Spring-Mass Response to a Suddenly Applied Load

- (7) Assume the loading  $F(t)$  is a suddenly applied load of magnitude  $F_0$ .
- (8) Find the particular solution for this case of a suddenly applied load.
- (9) Write the complete solution as the sum of  $u_c(t)$  from (6) and  $u_p(t)$  from (8).
- (10) Assume “at rest” initial conditions to find the constants  $A$  and  $B$  and write the complete solution  $u(t)$  as a function of  $F_0$ ,  $k$ , and  $\cos \omega_n t$ .
- (11) Use (10) to find the largest displacement  $u_{\max}$  and compare it with the static displacement  $u_{\text{st}}$ .

### C.6 Calculate Cantilever Response to Suddenly Applied Weight



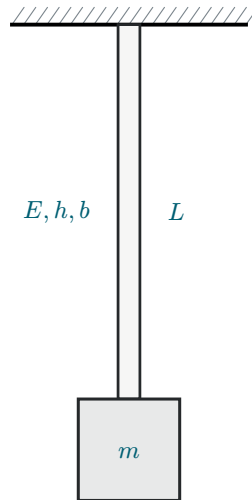
**Figure 18:** Cantilever beam with tip mass.

Take  $g = 9.81 \text{ m/s}^2$  as the generally accepted value of gravitational acceleration. Consider a cantilever beam of length  $L = 568 \text{ mm}$  and a mass  $m = 5.35 \text{ kg}$  that has to be attached to its tip as shown in Figure 18. The beam is made of steel with elastic modulus  $E = 200 \text{ GPa}$ , yield stress  $Y = 710 \text{ MPa}$ , and cross-section dimensions  $h = 3 \text{ mm}$  and  $b = 37 \text{ mm}$ .

**Do the following.**

- (1) Display input data.
- (2) Calculate the cross-section second moment of area  $I$ , the flexural stiffness  $EI$ , the spring constant  $k$ , the natural frequency  $\omega_n$  rad/s,  $f_n$  Hz, and period of oscillation  $\tau$  of the system.
- (3) Calculate the weight  $F_0$  of the mass  $m$ .
- (4) Static case. Assume the mass is attached to the undeformed beam and then released gradually without inducing vibration. Calculate the static displacement  $u_{\text{st}}$ .
- (5) Dynamic case. Assume the mass is attached to the undeformed beam and then released suddenly, inducing vibration.
  - (a) Calculate the maximum dynamic displacement  $u_{\text{dyn}}$ .
  - (b) Plot the response  $u(t)$  measured from the undeformed position for  $N_{\text{osc}} = 10$  oscillations.
  - (c) Place a data tip on the plot to identify the maximum displacement and compare it with the result of item (5)(a).
- (6) Graduate work. Calculate the maximum stress at the root of the beam and the safety factor  $\text{SF} = Y/\sigma$ , and comment on your results for the following two cases.
  - (a) Static-case stress  $\sigma_{\text{st}}$ .
  - (b) Dynamic-case stress  $\sigma_{\text{dyn}}$ .
- (7) Discuss your results.

### C.7 Analyze Flex-Beam Pendulum



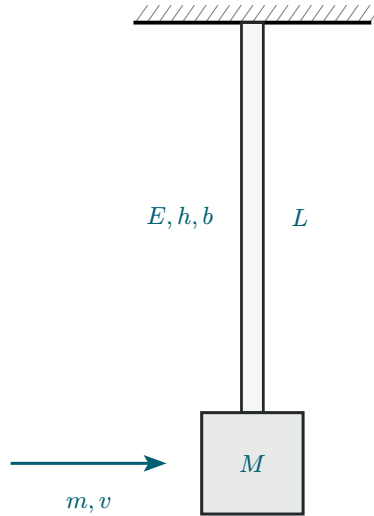
**Figure 19:** *Flex-beam pendulum.*

Consider a cantilever beam of length  $L$  with a mass  $m$  attached at its tip as shown in Figure 19. The beam is fixed into the ceiling and is hanging downward along the  $z$  axis. The mass may oscillate sideways in either  $x$  or  $y$  directions. The beam cross-section properties are  $E, h, b$ .

**Do the following.**

- (1) Calculate the minor and major second moments of area  $I_1$  and  $I_2$ .
- (2) Assume a horizontal displacement  $u(t)$  which produces flexure about the minor  $I_1$ .
- (3) Draw free-body diagram (FBD).
- (4) Write Newton's law of motion (NLM) equation.
- (5) Apply small-angle approximation and deduce the equation of motion (EOM). Clearly state your assumptions.
- (6) Define the natural frequency  $\omega_{n,\text{FBP}}$  of the flex-beam pendulum.
- (7) Deduce the governing equation.
- (8) Compare your results with those of Problem B.1 and write the general solution.
- (9) Solve the initial value problem (IVP), i.e., deduce the response for simultaneous application of initial displacement  $u_0$  and initial velocity  $\dot{u}_0$ .
- (10) Graduate work. Explain how the solution would change if flexure were about  $I_2$ .

## C.8 Calculate Flex-Beam Pendulum Response to Impact



**Figure 20:** Impact excitation of a vertical flex-beam pendulum.

Take  $g = 9.81 \text{ m/s}^2$  as the generally accepted value of gravitational acceleration. Consider a vertical flex-beam pendulum of length  $L = 568 \text{ mm}$  with a mass  $M = 42 \text{ kg}$  attached at its end, as shown in Figure 20. The beam is steel with  $E = 200 \text{ GPa}$  and cross-section dimensions  $h = 3 \text{ mm}$  and  $b = 37 \text{ mm}$ . The beam is initially at rest. A projectile of mass  $m = 18 \text{ g}$  and velocity  $v = 935 \text{ m/s}$  hits the mass and remains embedded in it. The projectile orientation causes the beam to flex about its minor cross-sectional axis. The impact imparts an initial velocity to the system, which then oscillates.

**Do the following.**

- (1) Display input data.
- (2) Calculate the minor second moment of area  $I$  of the beam.
- (3) Calculate the minor cross-sectional stiffness  $EI$ .
- (4) Calculate the flex-beam spring constant  $k$ .
- (5) Calculate the initial velocity of the system  $\dot{u}_0$  as resulting from the interaction between the projectile and the large mass.
- (6) Calculate the rad/s natural frequency  $\omega_{n,\text{FBP}}$  and period of oscillation  $\tau_{\text{FBP}}$  of the flex-beam pendulum with projectile embedded into it.
- (7) Calculate the response  $u(t_1)$  after  $t_1 = 10 \text{ s}$ .
- (8) Plot the response of the system for  $N_{\text{osc}} = 10$  oscillations and find on the plot:
  - (a) The largest response.
  - (b) The response value calculated at item (7).
- (9) Compare your results with those of Problem A.3.

### C.9 Complex Exponentials and Trigonometric Functions

- (1) Find the solutions of  $z^4 = 1$  (the fourth roots of unity) and express them as:
  - (a) Complex numbers.
  - (b) Complex exponentials.
- (2) Write the complex number  $e^{i\omega t}$  as a sum of trigonometric functions (hint, Euler's formula).
- (3) Calculate the phase difference  $\phi$  in radians and degrees between the following signals:
  - (a) Compare  $x_1(t) = \sin(\omega t)$  and  $x_2(t) = \sin(\omega t + 3\pi/4)$ .
  - (b) Compare  $x_1(t) = \sin(\omega t + \pi/3)$  and  $x_2(t) = \sin(\omega t + 3\pi/4)$ .
  - (c) Compare  $x_1(t) = \sin(\omega t + 45^\circ)$  and  $x_2(t) = \sin(\omega t + 115^\circ)$ .
- (4) What is the phase in radians and degrees of the following complex numbers:
  - (a)  $e^{i\pi/2}$ .
  - (b)  $e^{i3\pi/2}$ .
  - (c)  $e^{i\pi/4}$ .
  - (d)  $e^{-i\pi/2}$ .
  - (e)  $e^{-i\pi/4}$ .
- (5) Draw the locations on the unit circle of the complex numbers listed in item (4).
- (6) Calculate the magnitude and phase in degrees of the complex number  $Z = 4 + 5i$ .